

# AI FAQ Assistant API

## 6. Project Testing Phase

### 1. Introduction

This document describes the testing activities carried out on the AI FAQ Assistant API to verify that it meets the functional and non-functional requirements defined earlier, and that it behaves correctly, securely, and reliably before deployment.

### 2. Testing Objectives

- Verify that all functional requirements (FR-1 through FR-12) are correctly implemented
- Confirm that the NLP matching engine returns accurate answers, including for paraphrased questions
- Ensure admin-only endpoints are properly protected by authentication
- Confirm the system responds within acceptable time limits under normal load
- Identify and resolve defects before deployment sign-off

### 3. Testing Scope

Testing covers the question-answering endpoint, FAQ management endpoints, authentication, feedback capture, and basic analytics retrieval. Load testing is performed at a scale appropriate to a pilot deployment rather than full production traffic.

### 4. Types of Testing Performed

| Test Type           | Purpose                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------|
| Unit Testing        | Verify individual functions (e.g., matching logic, CRUD operations) work correctly in isolation |
| Integration Testing | Verify that the API, database, and NLP engine work together correctly                           |
| System Testing      | Verify the complete system meets requirements end-to-end                                        |

|                               |                                                                             |
|-------------------------------|-----------------------------------------------------------------------------|
| Performance Testing           | Measure response time under normal and moderately increased load            |
| Security Testing              | Verify authentication, authorization, and input validation behave correctly |
| User Acceptance Testing (UAT) | Confirm the system meets user and business expectations before sign-off     |
| Regression Testing            | Re-test previously working features after bug fixes to ensure nothing broke |

## 5. Test Environment

- Dedicated staging database populated with a representative sample FAQ dataset
- Postman collection covering all documented API endpoints
- Separate admin and non-admin test accounts to verify access control

## 6. Sample Test Cases

| Test ID | Scenario                                                    | Expected Result                                    | Requirement Traced |
|---------|-------------------------------------------------------------|----------------------------------------------------|--------------------|
| TC-01   | Ask a clearly worded FAQ question                           | Correct answer returned with high confidence score | FR-1, FR-2, FR-3   |
| TC-02   | Ask a paraphrased/reworded version of a stored FAQ question | Correct FAQ still matched via NLP similarity       | FR-2               |
| TC-03   | Ask an unrelated or nonsense question                       | Fallback 'no match found' response returned        | FR-4               |
| TC-04   | Admin creates a new FAQ with valid data                     | FAQ saved and retrievable via the API              | FR-5               |
| TC-05   | Admin updates an existing FAQ                               | Updated content is reflected in subsequent queries | FR-6               |
| TC-06   | Admin deletes an FAQ                                        | FAQ no longer returned by /api/ask or /api/faqs    | FR-7               |

|       |                                                     |                                                       |                             |
|-------|-----------------------------------------------------|-------------------------------------------------------|-----------------------------|
| TC-07 | Non-admin attempts to access an admin-only endpoint | Request rejected with an authorization error          | FR-11                       |
| TC-08 | Submit feedback on a returned answer                | Feedback recorded and linked to the correct query log | FR-10                       |
| TC-09 | Send several concurrent requests to /api/ask        | All requests handled within acceptable response time  | Non-functional: Performance |
| TC-10 | Submit an empty or malformed question               | System returns a clear validation error, not a crash  | Non-functional: Reliability |

## 7. Bug Tracking Summary

| Bug ID | Description                                                      | Severity | Status                                        |
|--------|------------------------------------------------------------------|----------|-----------------------------------------------|
| BUG-01 | Low-confidence matches were sometimes returned as valid answers  | Medium   | Fixed — confidence threshold adjusted         |
| BUG-02 | Missing input validation allowed empty questions to be submitted | Low      | Fixed — validation added                      |
| BUG-03 | Admin authentication token was not expiring correctly            | High     | Fixed — token expiry logic corrected          |
| BUG-04 | Deleted FAQs still appeared in analytics reports                 | Low      | Fixed — soft-delete flag respected in queries |

## 8. Test Results Summary

All planned test cases across unit, integration, system, performance, security, and acceptance testing categories were executed. The core functional requirements — question matching, FAQ

management, authentication, and feedback logging — passed successfully. All identified defects were fixed and re-verified through regression testing prior to final sign-off.

| Test Category       | Total Cases | Passed | Failed (before fix) |
|---------------------|-------------|--------|---------------------|
| Unit Testing        | 18          | 18     | 2                   |
| Integration Testing | 12          | 12     | 1                   |
| System Testing      | 10          | 10     | 1                   |
| Security Testing    | 6           | 6      | 1                   |
| UAT                 | 8           | 8      | 0                   |

## 9. Performance Testing Observations

Under simulated moderate load, the /api/ask endpoint responded within the target range of 1–2 seconds for the majority of requests, meeting the non-functional performance requirement defined during Requirement Analysis. Response times remained stable as the FAQ dataset size increased, confirming that pre-computing embeddings at creation time was an effective design decision.

## 10. Deployment Readiness Checklist

- All critical and high-severity bugs resolved and verified
- API documentation (Swagger/OpenAPI) finalized and matches actual endpoint behavior
- Performance within acceptable response-time limits under test load
- Security checks — authentication, authorization, and input validation — passed
- Backup of the staging database taken before final deployment
- Sign-off obtained from the project team confirming readiness for deployment

## 11. Tools Used for Testing

| Tool                                   | Purpose                                  |
|----------------------------------------|------------------------------------------|
| Postman                                | Manual and scripted API endpoint testing |
| Jest / PyTest (as applicable to stack) | Automated unit and integration tests     |

|                             |                                                            |
|-----------------------------|------------------------------------------------------------|
| Swagger UI                  | Validating that documented endpoints match actual behavior |
| Load-testing script (basic) | Simulating concurrent requests for performance testing     |

## 12. Sample Detailed Test Case

| Field           | Details                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| Test ID         | TC-02                                                                                                     |
| Title           | Verify semantic matching handles paraphrased questions                                                    |
| Precondition    | FAQ 'How do I reset my password?' exists in the database                                                  |
| Steps           | 1) Send POST /api/ask with question 'I forgot my password, what do I do?' 2) Inspect the response         |
| Expected Result | The response returns the password-reset FAQ answer with a confidence score above the configured threshold |
| Actual Result   | Matched correctly; confidence score 0.88                                                                  |
| Status          | Passed                                                                                                    |

## 13. Regression Testing Summary

After each bug fix listed in Section 7, the related feature area and its immediate neighbors were re-tested to confirm no new issues were introduced. For example, after fixing BUG-03 (token expiry), the full authentication flow — login, token use, and token expiry — was re-verified end-to-end rather than testing the fix in isolation.

## 14. Lessons Learned from Testing

- Testing with paraphrased, real-world-style questions (not just exact FAQ text) was essential to properly validate the NLP matching feature
- Writing test cases early, alongside development, made it easier to catch issues like BUG-01 sooner rather than later
- Clear requirement traceability (Section 6) made it straightforward to confirm that every requirement had at least one corresponding test case

## 15. Sign-off Summary

| Item                                                                           | Status    |
|--------------------------------------------------------------------------------|-----------|
| All Must-have functional requirements tested                                   | Complete  |
| All identified defects resolved and re-verified                                | Complete  |
| Performance within target response time                                        | Confirmed |
| Security checks passed                                                         | Confirmed |
| Documentation finalized (Requirements, Design, Planning, Development, Testing) | Complete  |

## 16. Post-Deployment Monitoring Plan

After deployment, the team plans to continue monitoring the `query_logs` table for low-confidence or unanswered questions, using this data to periodically expand and refine the FAQ dataset. This closes the loop back to the original objective from the Ideation phase — continuously improving accuracy using logged queries and feedback.

## 17. Detailed Walkthrough of a Security Test

To illustrate the security testing approach in more depth, this section walks through the specific steps used to verify that admin-only endpoints are properly protected, corresponding to TC-07 in Section 6.

- 1. A test account without admin privileges is used to obtain a valid but non-admin authentication token.
- 2. A request is sent to an admin-only endpoint (e.g., `POST /api/faqs`) using this non-admin token.
- 3. The system is expected to reject the request with an appropriate authorization error rather than processing it.
- 4. The same request is then repeated with no token at all, and separately with a deliberately malformed token, to confirm both cases are also correctly rejected.
- 5. Finally, a valid admin token is used to confirm the same endpoint succeeds as expected, ensuring the protection does not block legitimate admin use.

This layered approach — testing the failure cases as well as the success case — gives higher confidence that the authentication design from the Design phase (Section 10, Security Design Notes) is correctly implemented, rather than only confirming that the 'happy path' works.

## 18. Overall Quality Assessment

Taken together, the results in Sections 6 through 9 indicate that the AI FAQ Assistant API is functionally complete against its Must-have requirements, reasonably accurate on paraphrased questions, secure against unauthorized admin access, and performant enough for its intended pilot scale. Remaining minor issues identified during testing were either resolved (see Section 7) or noted as low-priority polish items that do not block deployment.

## 19. Recommendations for Future Test Cycles

- Expand the paraphrase test set over time using real questions captured from `query_logs`, rather than relying only on manually written test questions
- Introduce automated regression test suites so that future feature additions can be verified quickly without repeating all manual steps
- Periodically re-run performance testing as the FAQ dataset grows, to confirm response times remain within target

## 20. Conclusion

Testing confirmed that the AI FAQ Assistant API meets its documented functional and non-functional requirements. With all identified defects resolved and verified, the system is considered ready for deployment, completing the project lifecycle from initial brainstorming through to a tested, working API.